Definitions and Examples

Supremum and Infimum. For a set $S \subset \mathbb{R}$:

- The *supremum* (least upper bound) is the smallest $M \in \mathbb{R}$ such that $x \leq M$ for all $x \in S$. Denoted $\sup S$.
- The *infimum* (greatest lower bound) is the largest $m \in \mathbb{R}$ such that $m \leq x$ for all $x \in S$. Denoted $\inf S$.

Example. For $S = (0,1)$, $\sup S = 1$, $\inf S = 0$.

Partitions, Upper and Lower Sums. A *partition* of $[a,b]$ is a finite set

$$P = \{x_0, x_1, \ldots, x_n\}, \quad a = x_0 < x_1 < \cdots < x_n = b.$$

For a bounded function $f : [a,b] \to \mathbb{R}$:

$$U(f,P) = \sum_{i=1}^{n} M_i(x_i - x_{i-1}), \quad M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\},$$

$$L(f,P) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}), \quad m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}.$$

Example. For $f(x) = x$ on $[0,1]$ with $P = \{0, 0.5, 1\}$:

$$U(f,P) = 0.75, \quad L(f,P) = 0.25.$$

Integrability. A bounded $f : [a,b] \to \mathbb{R}$ is *Riemann integrable* if

$$\sup_P L(f,P) = \inf_P U(f,P).$$

Example. $f(x) = x$ on $[0,1]$ is integrable with $\int_0^1 x dx = \frac{1}{2}$.

Characteristic Function. For $A \subset \mathbb{R}$, the *characteristic function* is

$$\chi_A(x) = \begin{cases} 1 & x \in A, \\ 0 & x \notin A. \end{cases}$$

Example. If $A = [0, \frac{1}{2}]$, then $\int_0^1 \chi_A(x) dx = \frac{1}{2}$.

Solutions

Exercise 4.1

Solution.

a) $\sup_{x \in A}(-f(x)) = -\inf_{x \in A} f(x)$.

b) $\inf_{x \in A}(-f(x)) = -\sup_{x \in A} f(x)$.

c) $\sup_{x\in A} \lambda f(x) = \lambda \sup_{x\in A} f(x)$ for $\lambda \geq 0$.

d) $\sup_{x\in A}(f(x) + g(x)) \leq \sup f + \sup g$.

e) $\inf_{x\in A}(f(x) + g(x)) \geq \inf f + \inf g$.

f) $\inf_{x\in A} \lambda f(x) = \lambda \inf f(x)$.

g) $|\sup f| \leq \sup |f|$.

h) $|\inf f| \leq \sup |f|$.

Exercise 4.2

Solution.

a) $U(-f, P) = -L(f, P)$.

b) $L(-f, P) = -U(f, P)$.

c) $U(\lambda f, P) = \lambda U(f, P)$ for $\lambda \geq 0$.

d) $L(\lambda f, P) = \lambda L(f, P)$.

e) $U(f + g, P) \leq U(f, P) + U(g, P)$.

f) $L(f + g, P) \geq L(f, P) + L(g, P)$.

Exercise 4.3

Solution. If $P \preceq Q$, then $U(f, Q) \leq U(f, P)$ and $L(f, Q) \geq L(f, P)$. Hence

$$0 \leq U(f, Q) - L(f, Q) \leq U(f, P) - L(f, P).$$

Exercise 4.4

Solution. The function χ_A is Riemann integrable since it has only finitely many discontinuities. Moreover,

$$\int_{-1}^{1} \chi_A(x)\, dx = 0,$$

because a finite set has measure zero.